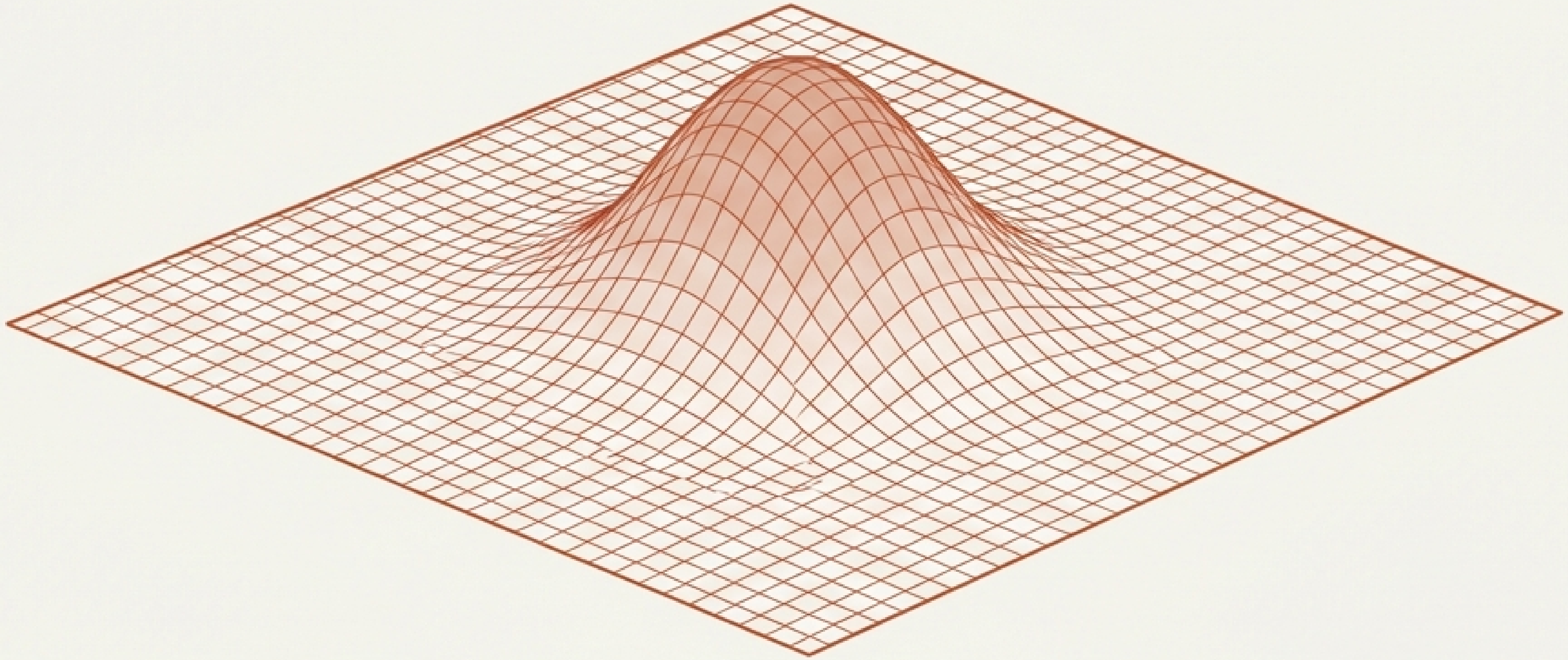
Fermion Masses as Localized Energy Functionals

A Geometric Interpretation of Higgs–Lepton Couplings & The Origin of Mass Hierarchy



Abstract: A proposal where fermion masses arise from stable configurations, and Higgs couplings are identified with scalar susceptibilities.

Based on the work of Omar Iskandarani (Independent Researcher, Groningen)

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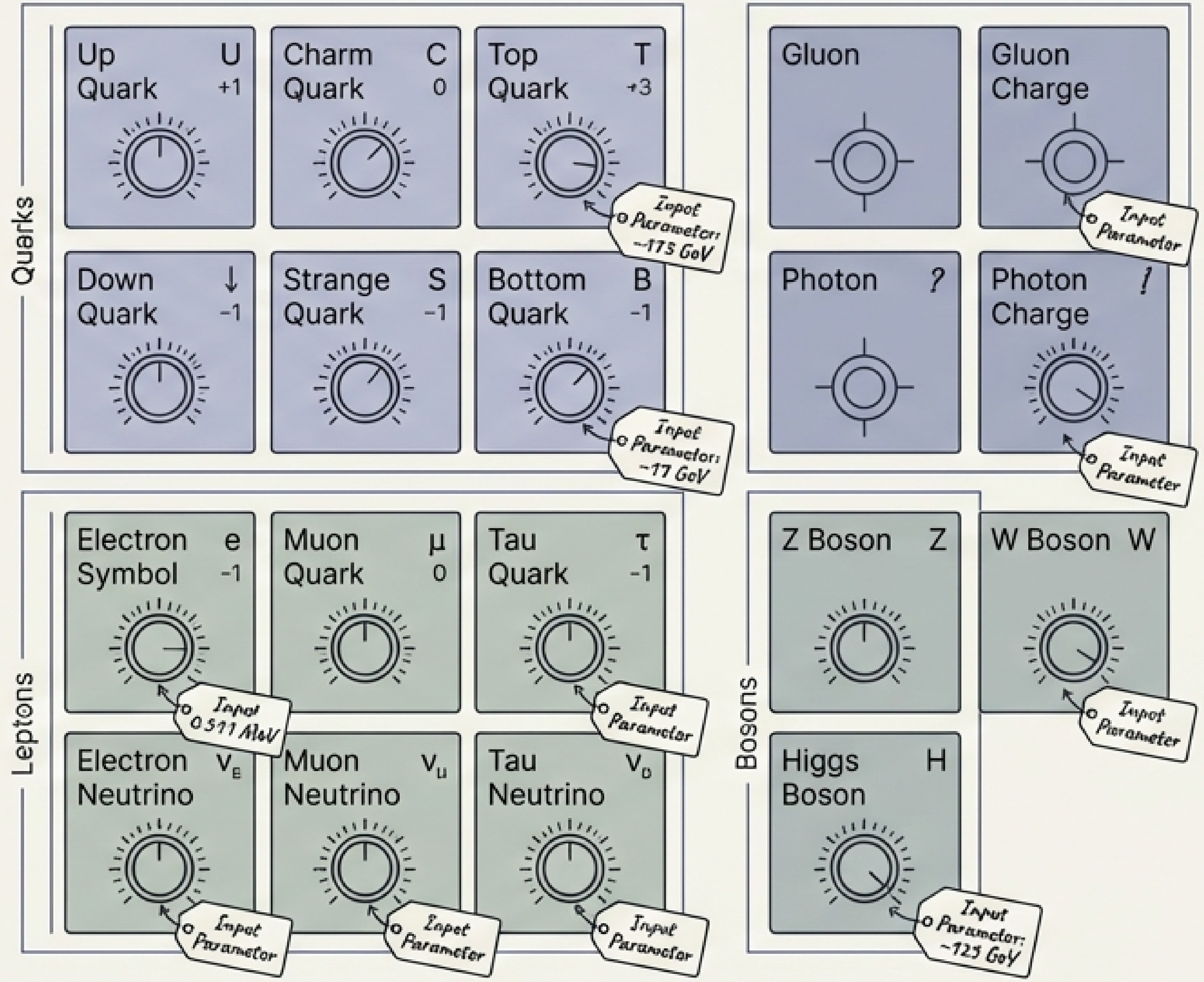
The Standard Model is complete, yet structurally silent.

The discovery of the Higgs boson completed the SM particle content and confirmed mass generation via Yukawa interactions. However, the Yukawa sector remains phenomenological.

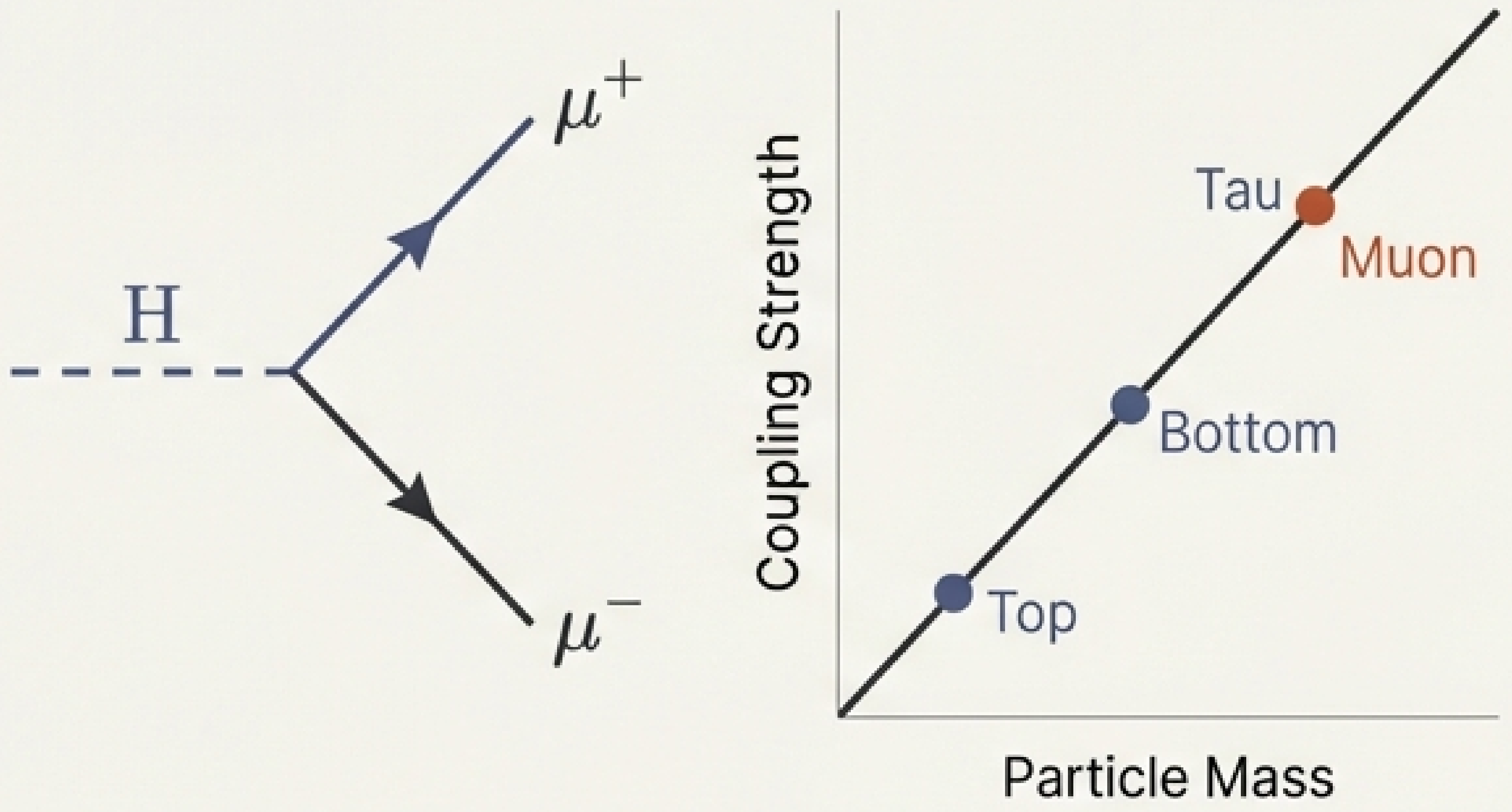
The Problem: We measure couplings, but we cannot explain them. Fermion masses span orders of magnitude without an underlying structural explanation.



Probing the Second Generation: $H \to \mu^+\mu^-$

Recent observation of the Higgs decay into a muon pair provides the first direct probe of couplings beyond the heavy 3rd generation. The measured rate is compatible with Standard Model expectations, reinforcing reinforcing the approximate proportionality: Coupling strength scales linearly with mass.

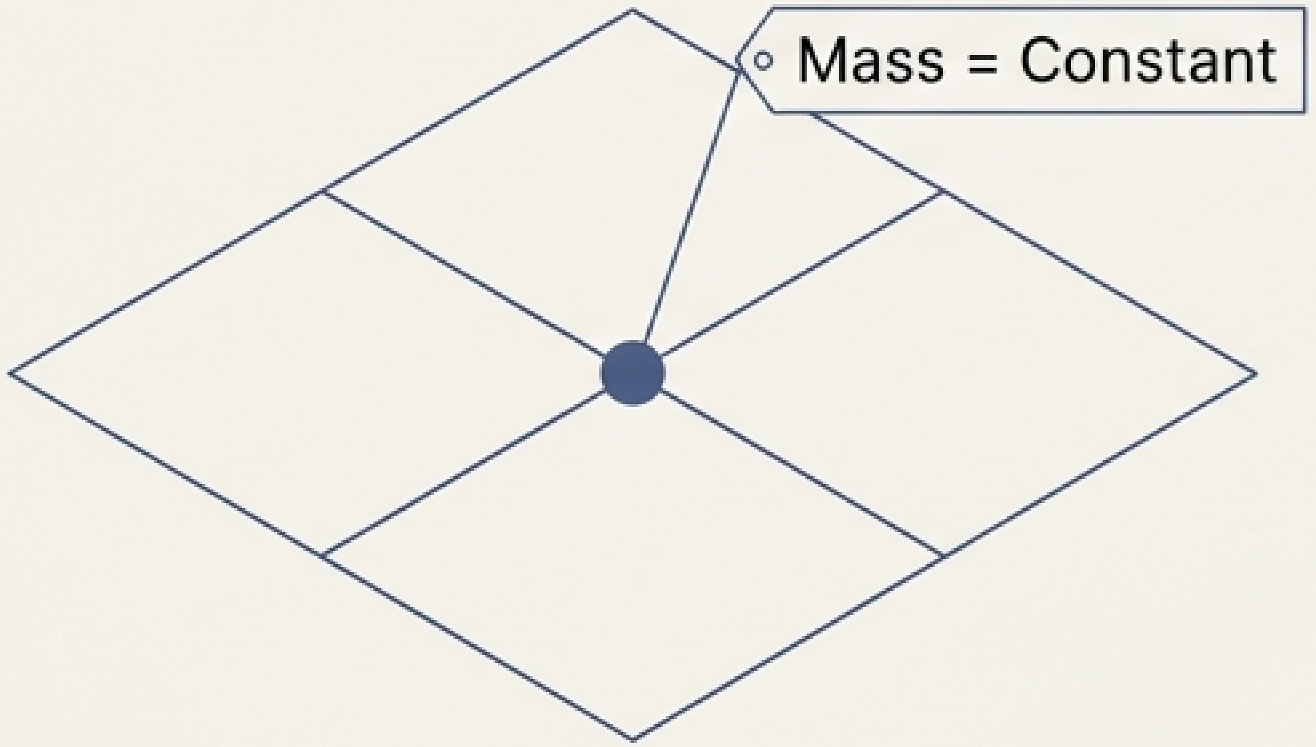
$$y \propto M$$



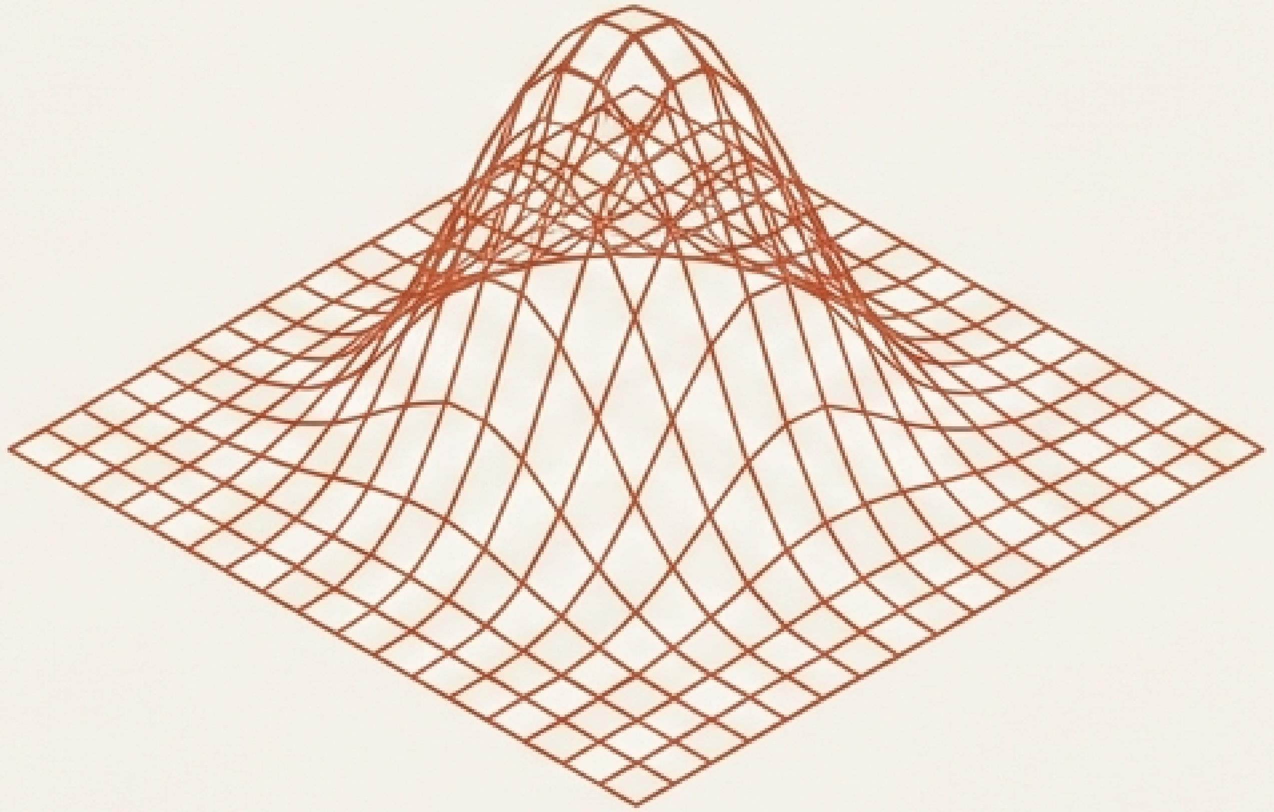
The Pivot: This confirms the relationship, but invites us to question the origin. Can we derive these constants from primitive, universal quantities?

Shifting Perspective: From Constants to Configurations

Standard View (Input Parameters)



Proposed View (Geometric Functional)



Concept: Fermion masses are modeled as localized energy functionals of stable configurations. Higgs couplings are identified with the response of these energies to a background field.

Constraint: Purely effective. No new particles. No new interactions.

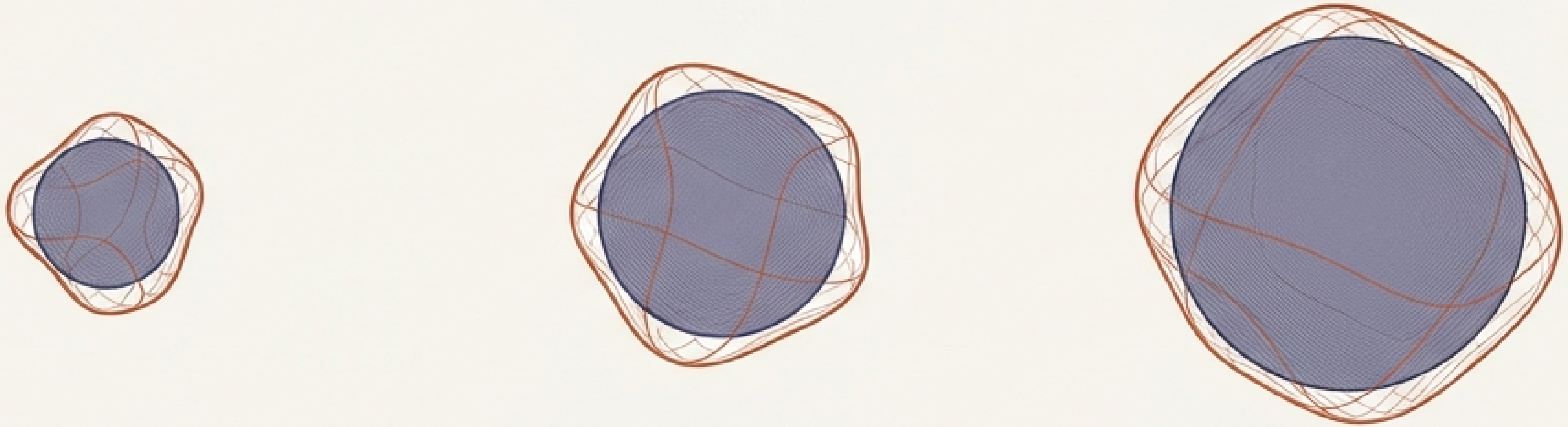
Defining Mass as Stored Energy

$$M_f = \mathcal{K}\,\mathcal{G}_f$$

Rest Energy
of the Fermion

Universal
Energy-Density Scale
(Constant for all)

Geometric Factor
(Unique to particle
shape/volume)

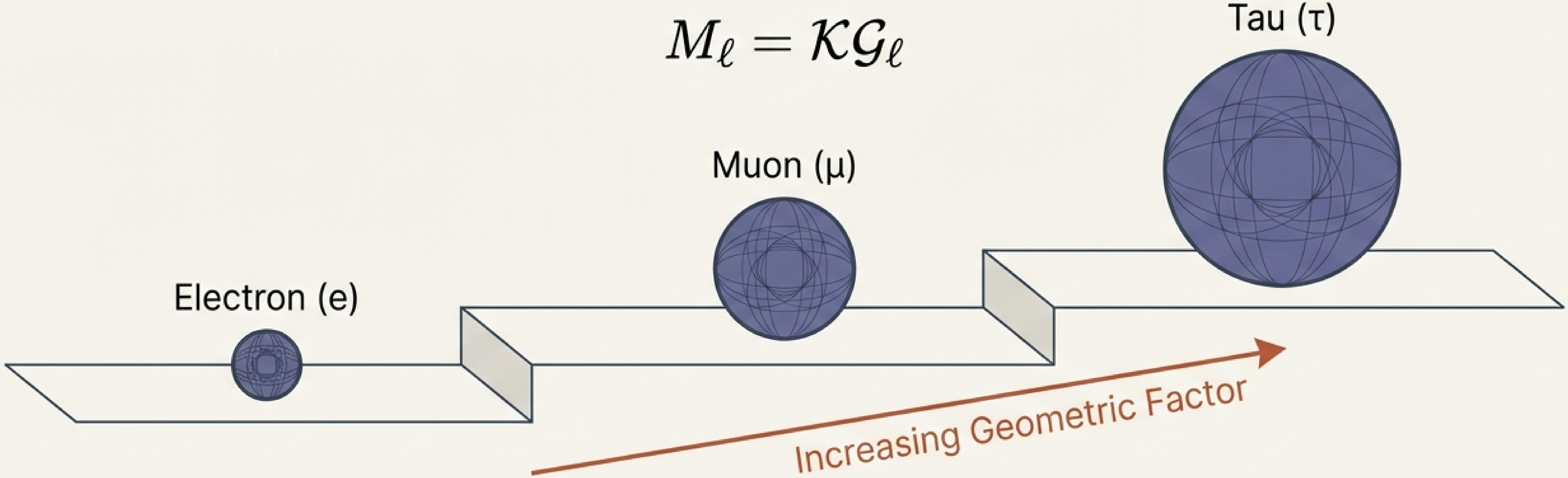


Mass is simply the total energy stored in a localized configuration.

Hierarchy is a Consequence of Geometry

Applying the mass definition to charged leptons (e, μ, τ):

$$M_\ell = \mathcal{K}\mathcal{G}_\ell$$



$\mathcal{G}_e$ (Small Support Volume) $\mathcal{G}_\mu$ (Medium Support Volume) $\mathcal{G}_\tau$ (Large Support Volume)

The massive difference between an electron and a tau is not a difference in fundamental nature, but a difference in **effective support volume** or internal coherence.

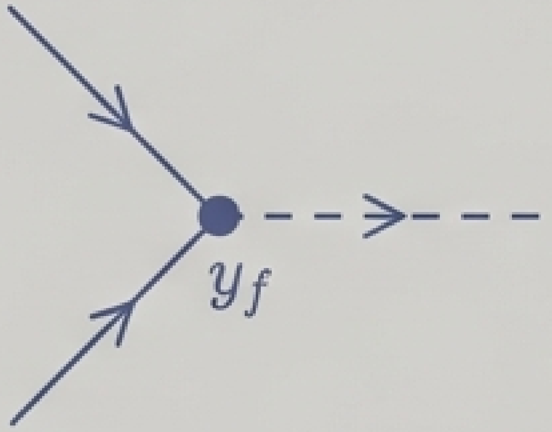
$$\mathcal{G}_e \ll \mathcal{G}_\mu \ll \mathcal{G}_\tau$$

Reinterpreting Coupling as Scalar Susceptibility

Standard Model View

$$L_Y = -y_f \bar{\psi}\psi H$$

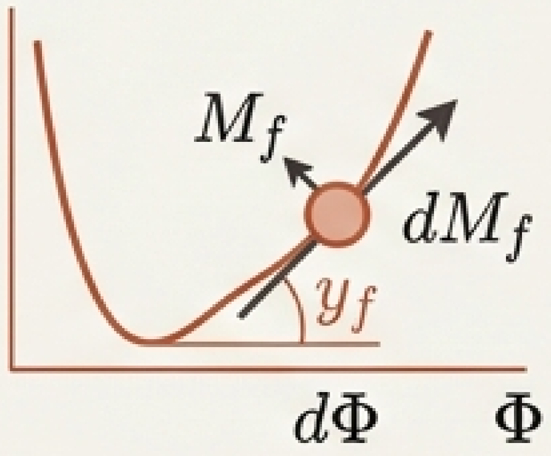
y_f is an arbitrary interaction constant.



Susceptibility View

$$y_f \equiv \frac{\partial M_f}{\partial \Phi}$$

Coupling is the response of the mass to the background field (Φ).



Analogy: Think of Yukawa couplings not as 'glues' of different strengths, but as 'susceptibilities'—how significantly the particle's energy state reacts to changes in the Higgs field.

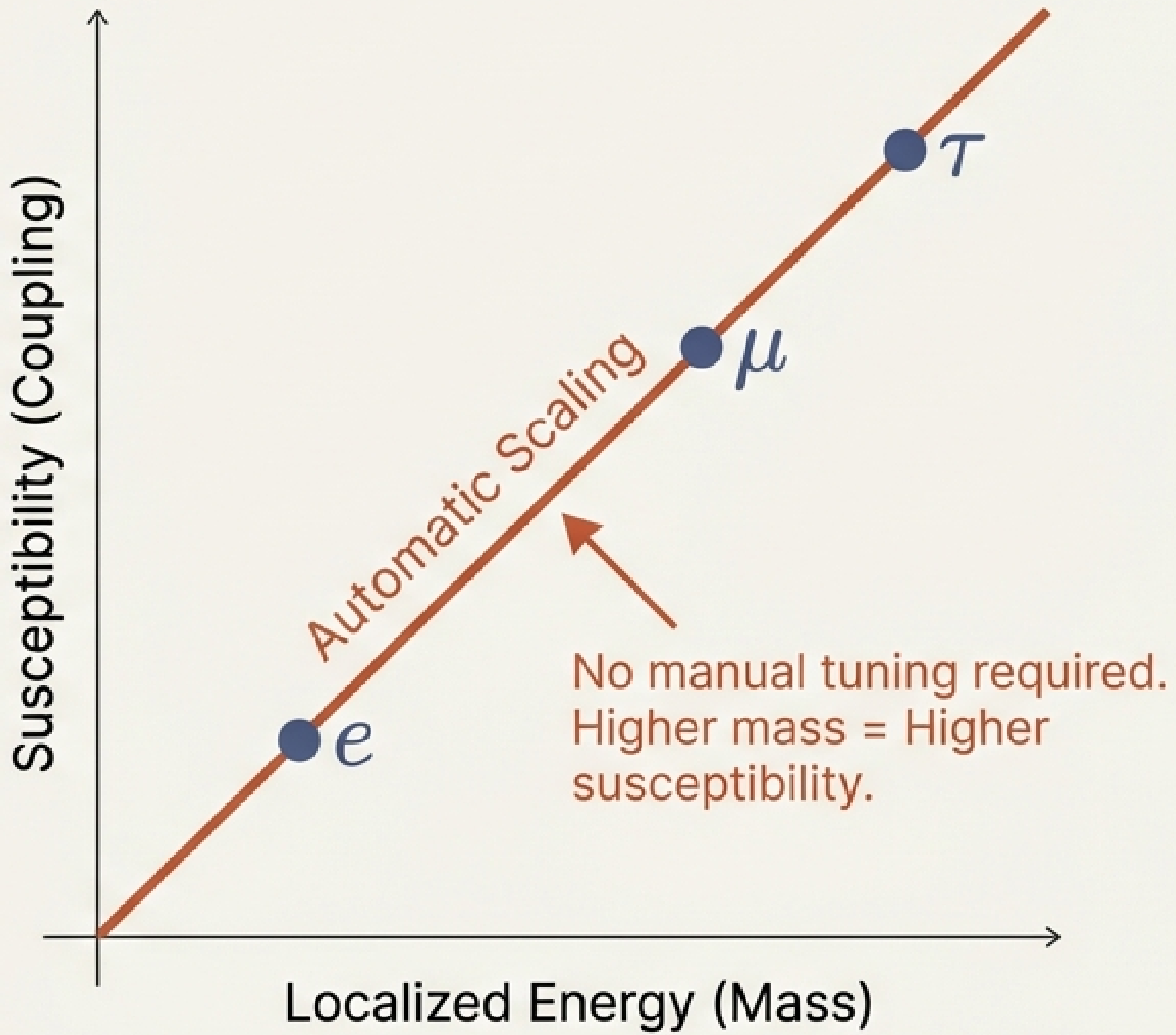
Proportionality is an Emergent Property

If the scalar field rescales the energy density:

$$\mathcal{E}_f(x) \to \mathcal{E}_f(x)[1 + \epsilon\Phi]$$

Then naturally:

$$y_f \propto M_f$$

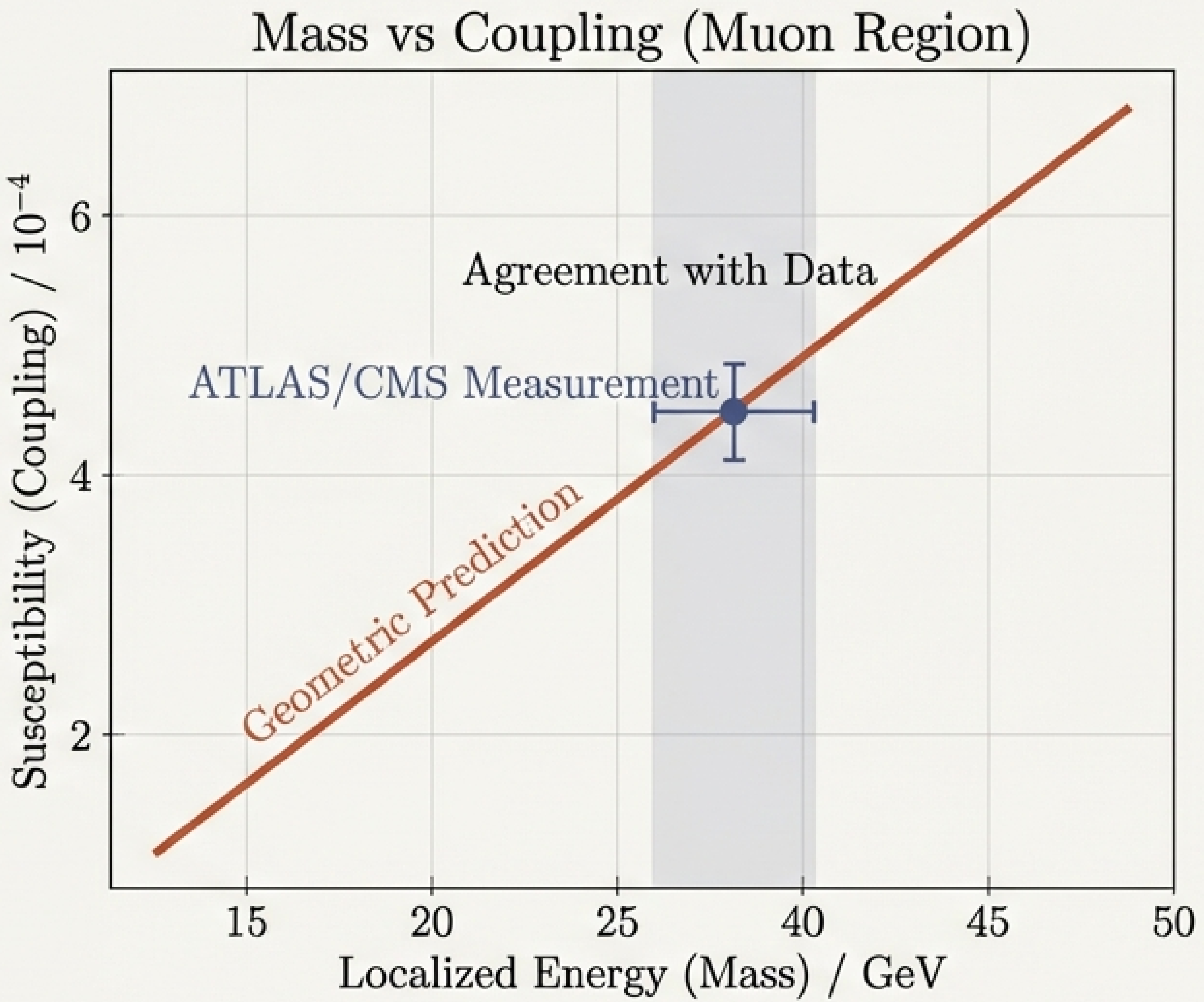


Consistency with $H \to \mu^+\mu^-$ Observation

The framework predicts that the "muonic configuration" must have a scalar susceptibility strictly proportional to its energy.

Experimental Result: Data confirms the Higgs-muon coupling fits the mass-proportional line. No anomalous enhancements found.

Conclusion: The theory survives the test of the second generation.

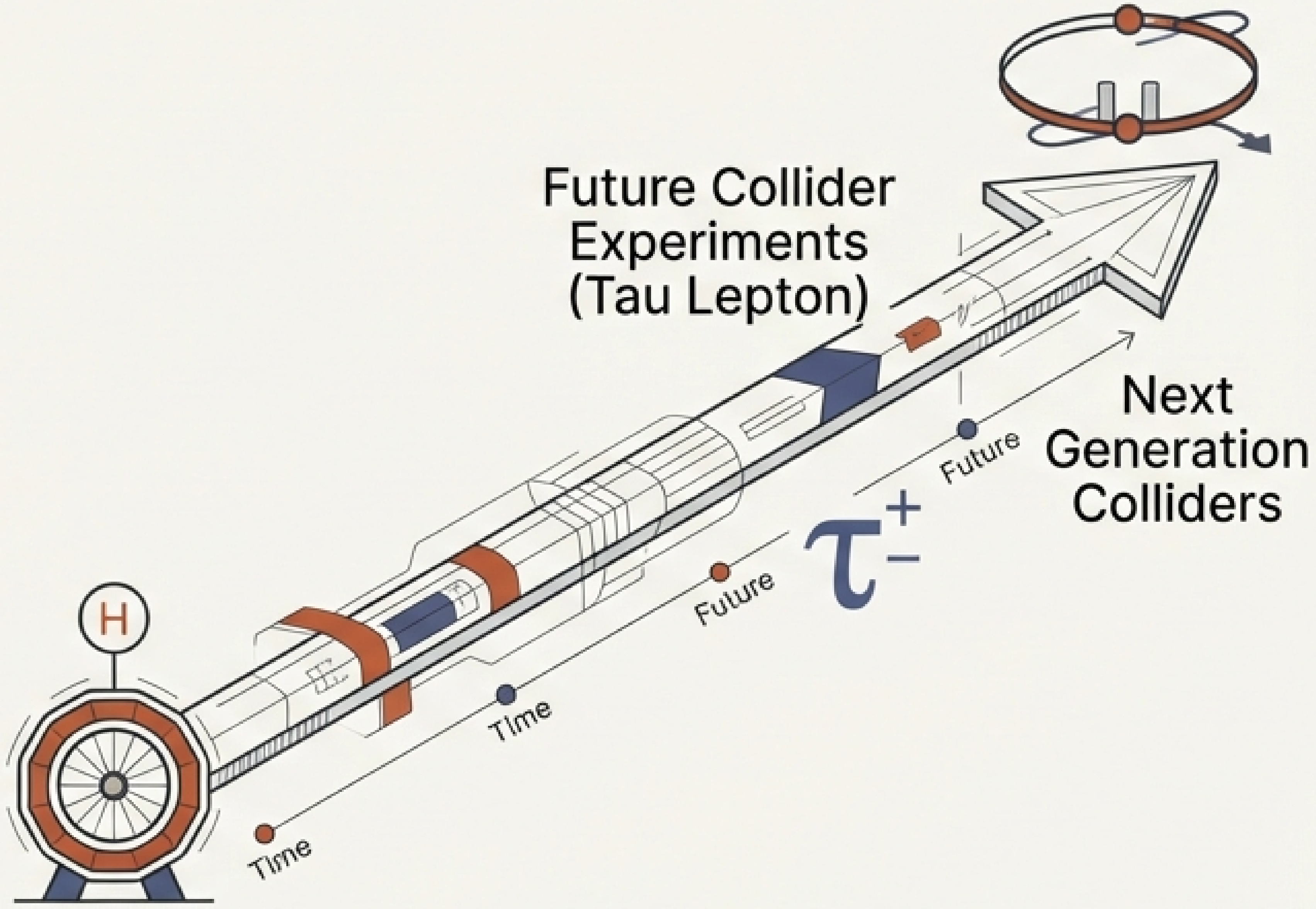


Predictions & Falsifiability

Prediction: Precision measurements of $H \to \tau^+\tau^-$ must continue to follow strict mass-proportional scaling.

The Constraint: Any significant deviation from linear scaling would signal a breakdown of this localization picture.

Significance: This implies scalar response is purely energy-dependent, not generation-dependent.



A Conservative, Effective Framework

Purely Effective – Does not require modifying the Standard Model Lagrangian.

No New Physics – No new particles or interactions are introduced.

Consistency – Remains fully consistent with the SM at presently accessible energies.

Value: It offers a complementary perspective on parameter structure, potentially guiding the search for a deeper underlying theory.

Summary: A Unified View of Mass and Coupling

Mass = Localized Energy Functional

$$M = \mathcal{K}\mathcal{G}$$

↓

Coupling = Response to Background Field

$$y = \partial M / \partial \Phi$$

↓

Hierarchy = Differences in Geometric Configuration

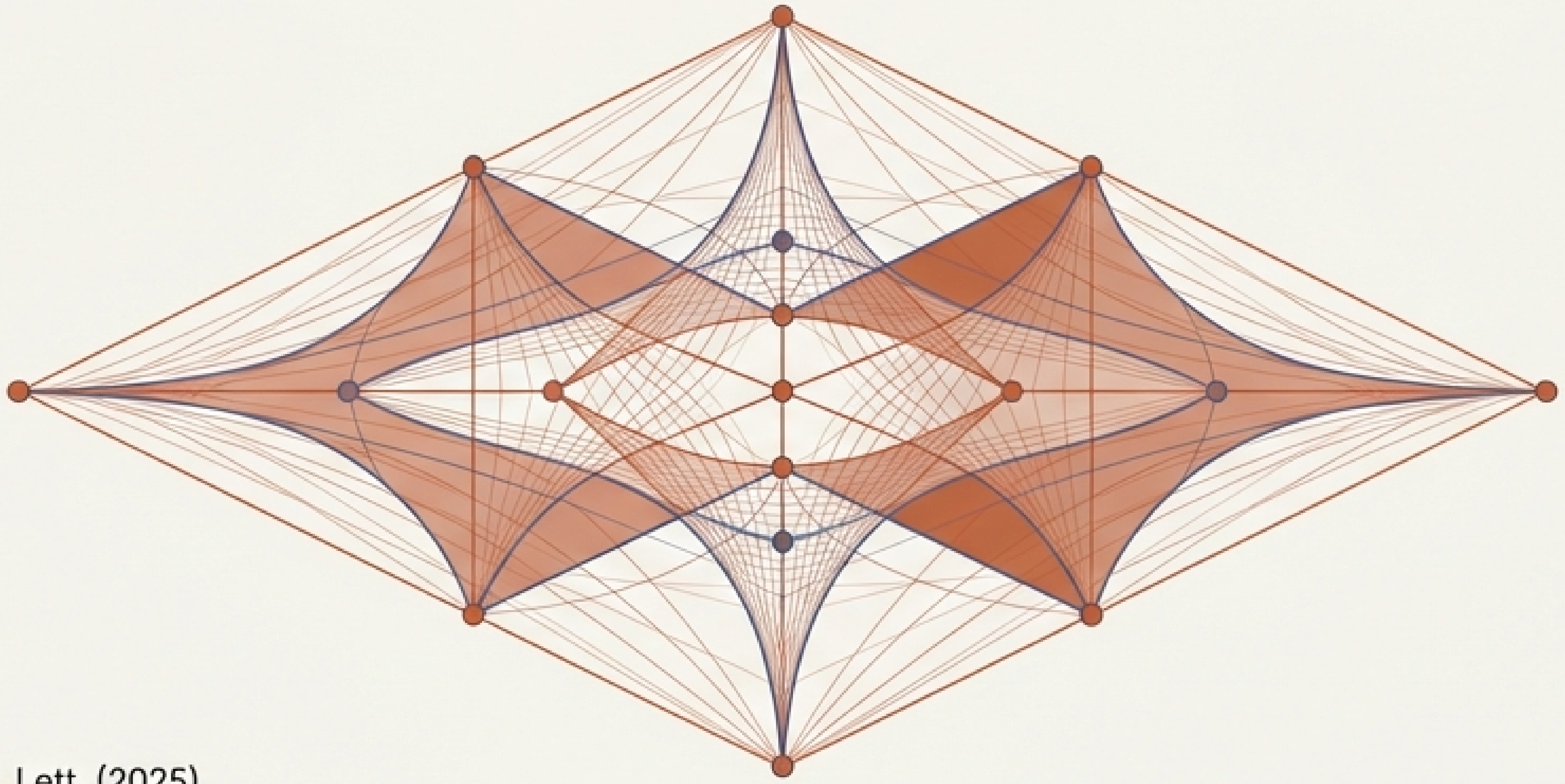
$$\mathcal{G}_f$$

↓

Observation: Consistent with Muon data; linear scaling is automatic.

Towards a Geometric Origin of Structure

By Viewing fermions through the lens of localized energy, the mysterious hierarchy of the Standard Standard Model transforms into a coherent geometric structure. The recent Higgs-muon results act as the first experimental anchor for this perspective beyond the third generation.



[1] ATLAS Collaboration, Phys. Rev. Lett. (2025).
[2] Particle Data Group, Review of Particle Physics.
[3] O. Iskandarani, Fermion Masses as Localized Energy Functionals (2026).